THE QUASI-PROJECTIVE SCHEMES OVER A BASE FORM A CATEGORY CLOSED UNDER BASE CHANGE, COMPOSITION, FIBRE PRODUCTS, SUMS, AND STANDARD SOURCES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13891-13914 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13916-13954, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to the already-collected projective-category card L16075; the placeholder list item is part of the pinned statement and was not silently dropped.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The quasi-projective five-way equivalence (lemma-quasi-projective, collected as L16072); the Morphisms-chapter base change and permanence lemmas for quasi-projective morphisms; the chapter's quasi-finite-separated-quasi-affine theorem (Zariski Main Type corollary).

Source reference. Original lemma 02LR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12426–12429. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B42: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13840–13853. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B3G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13954–13954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13954–13954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07RL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13954–13954. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B3H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13954–13954. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme which has an ample invertible sheaf. Let QP_S be the full subcategory of the category of schemes over S satisfying the equivalent conditions of Lemma 0B42.*

(1) *if $S' \to S$ is a morphism of schemes and S' has an ample invertible sheaf, then base change determines a functor $QP_S \to QP_{S'}$,*
(2) *if $X \in QP_S$ and $Y \in QP_X$, then $Y \in QP_S$,*
(3) *the category QP_S is closed under fibre products,*
(4) *the category QP_S is closed under finite disjoint unions,*
(5) *if $X \to S$ is projective, then $X \in QP_S$,*
(6) *if $X \to S$ is quasi-affine of finite type, then X is in QP_S,*
(7) *if $X \to S$ is quasi-finite and separated, then $X \in QP_S$,*
(8) *if $X \to S$ is a quasi-compact immersion, then $X \in QP_S$,*
(9) *add more here.*

Proof. Part (1) follows from Morphisms, Lemma 0B3G.

Part (2) follows from the fourth characterization of Lemma 0B42.

If $X \to S$ and $Y \to S$ are quasi-projective, then $X \times_S Y \to Y$ is quasi-projective by Morphisms, Lemma 0B3G. Hence (3) follows from (2).

If $X = Y \amalg Z$ is a disjoint union of schemes and $\mathcal{L}$ is an invertible $\mathcal{O}_X$-module such that $\mathcal{L}|_Y$ and $\mathcal{L}|_Z$ are ample, then $\mathcal{L}$ is ample (details omitted). Thus part (4) follows from the fourth characterization of Lemma 0B42.

Part (5) follows from Morphisms, Lemma 07RL.

Part (6) follows from Morphisms, Lemma 0B3H.

Part (7) follows from part (6) and Lemma 02LR.

Part (8) follows from part (7) and Morphisms, Lemma 01TN. □